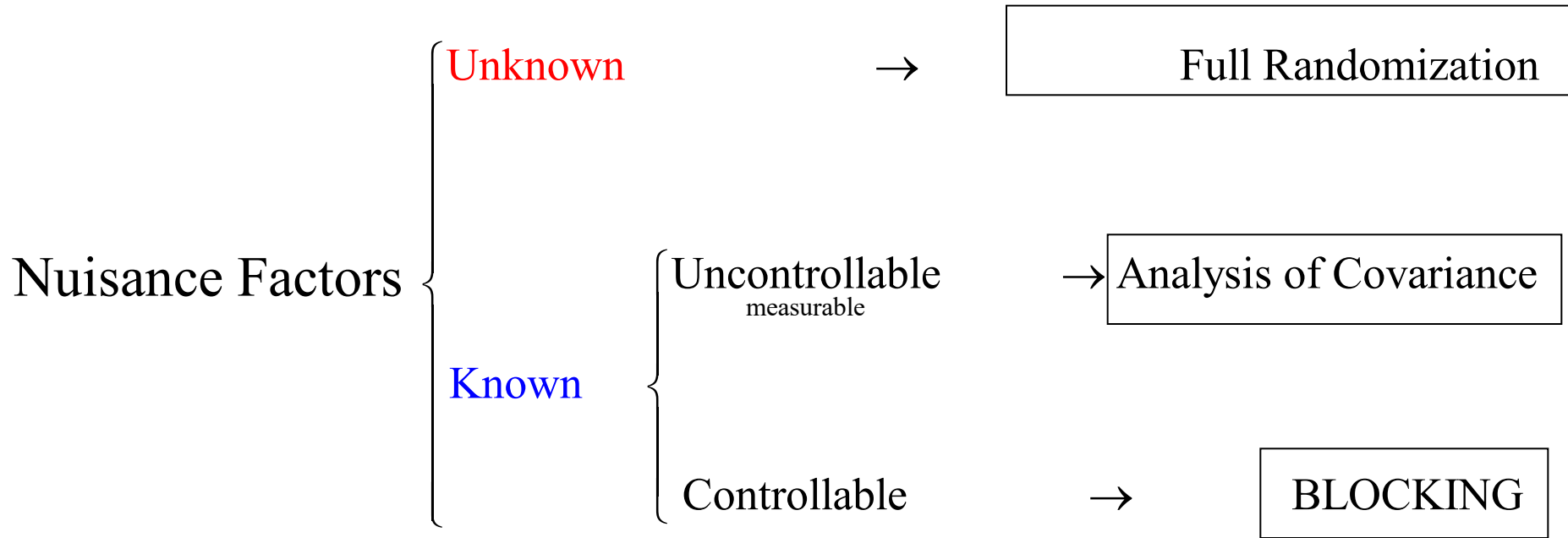


Experiments with Blocking Factors

Chapter 4

- A **Nuisance Factor (or Disturbing Factor)** is a factor that probably influences the Response Variable but it is of no interest for our purposes
- We want to **minimize the variability transmitted** to the Response Variable (captured by MS_E)
- Typical **Nuisance Factors** are:
 - Different material lots with identical nominal characteristics
 - Operators (with different skills)
 - Deterioration of fixtures/tools/machines or measuring instruments
 - Time (shift, week day, etc.)
- **Blocking** is an experimentation technique to manage known Nuisance Factors (consequences without blocking?)



‘Controllable’ means that we are able to set the levels of the Nuisance Factors

Block what you can and randomize what you cannot (R. Fisher)

Example: Tyre Brand Test

- A **car fleet manager** wants to establish if different Tyre Brands show different **Tread loss** after 20000Km.
- The manager is interested in finding out which **Tyre Brand out of four** (A;B;C;D) shows the minimum **Tread loss**.
- The manager design the experiment and he/she decides to perform the experiment using **5 sets of tyres** for each Brand

First Experimental Plan:
one car + complete randomization (in the table the run order

Brand			
A	B	C	D
1	2	19	3
9	6	11	13
7	16	8	4
15	14	10	5
12	18	20	17

Problems:

- Experimentation Time
- Results depending on the car and the driver

- **Second Experimental Plan: 4 different cars (1:4)**

Tyre Brand			
A	B	C	D
1	2	3	4
1	2	3	4
1	2	3	4
1	2	3	4
1	2	3	4

Experimentation Time → Reduced

But: In this experiment the tyre effect is completely confounded with the car effect

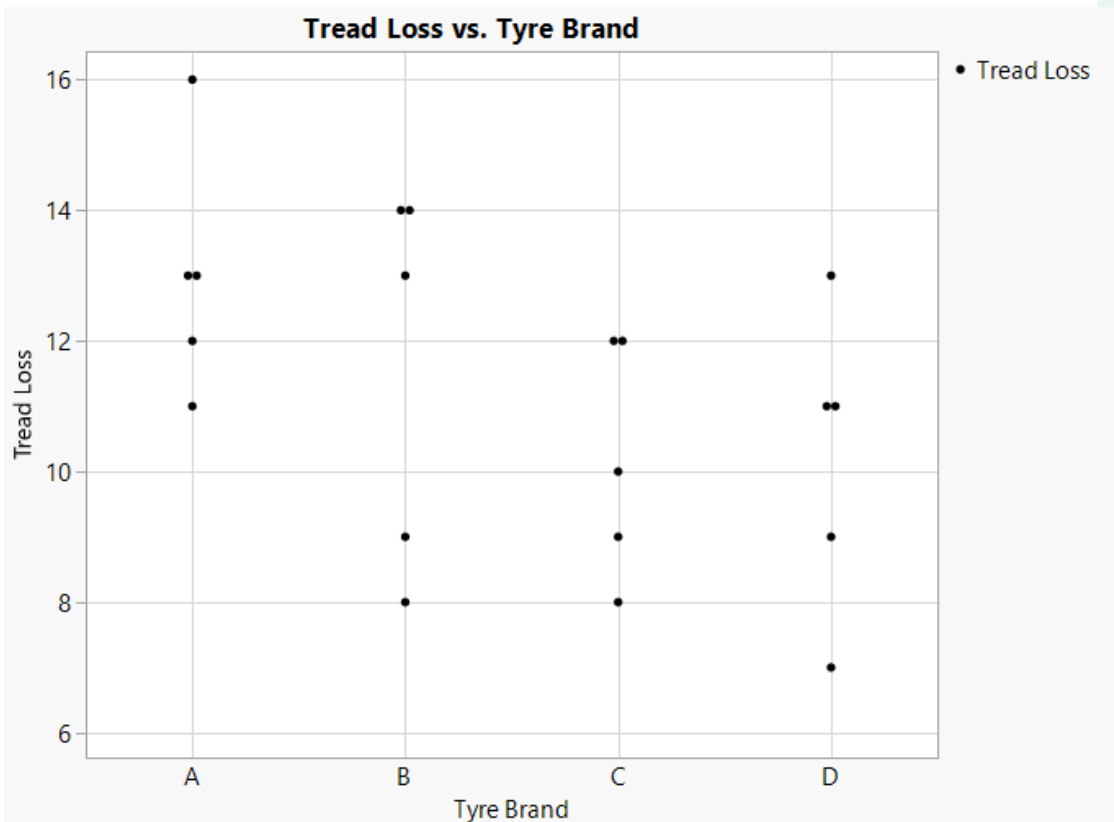
The manager changed the design by picking **5 cars**,
arranged the **experimental plan** and **run it randomly**

The results are in the table (the numbers are the tread loss in mm)

Blocking Factor-> Car

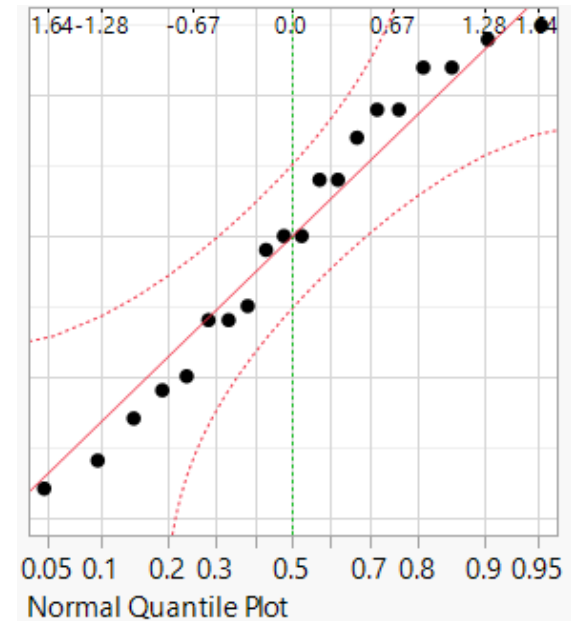
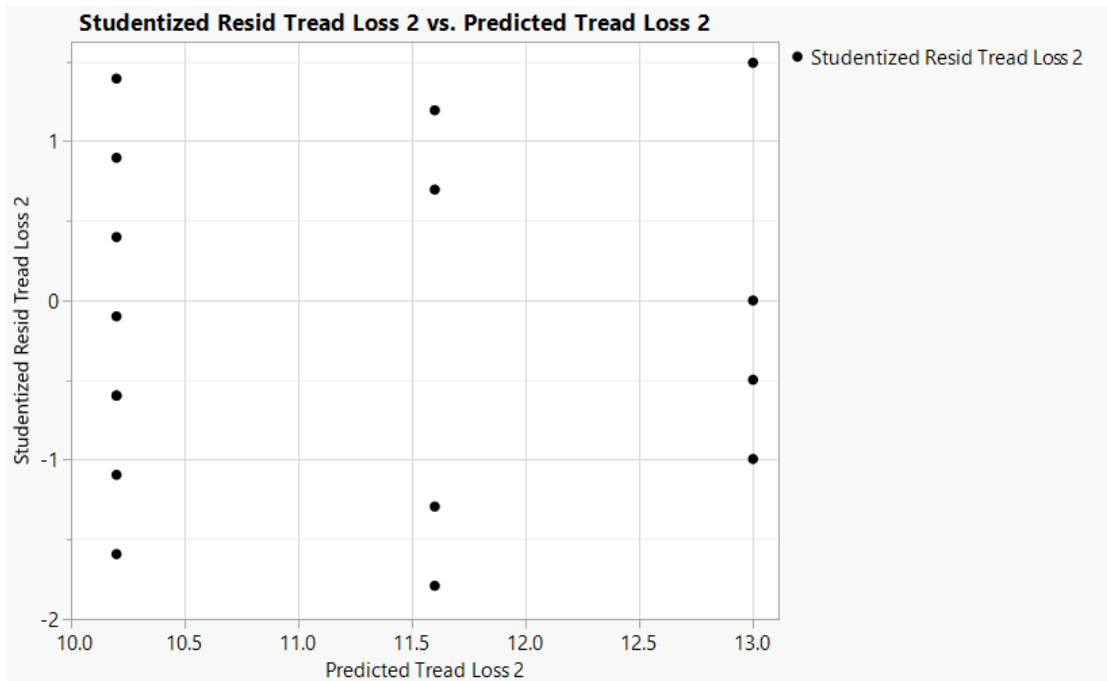
- **Data Plot**

	Tyre Brand			
Car	A	B	C	D
1	16	14	12	13
2	11	14	12	11
3	13	13	10	11
4	13	8	9	9
5	12	9	8	7



- Parameter estimation, SRES computation, Checking Hyp.

$$y_{ij} = \mu + \tau_i + \varepsilon$$



Goodness-of-Fit Test

	W	Prob<W
Shapiro-Wilk	0.9512923	0.3871
	A ²	Simulated p-Value
Anderson-Darling	0.3023199	0.5808

Note: Ho = The data is from the Normal distribution.
Small p-values reject Ho.

Test Brown-Forsythe
p_value=0.77

Conclusion about SRES?

- **OneWay ANOVA Table**

Summary of Fit

Rsquare	0.250116
Adj Rsquare	0.109513
Root Mean Square Error	2.247221
Mean of Response	11.25
Observations (or Sum Wgts)	20

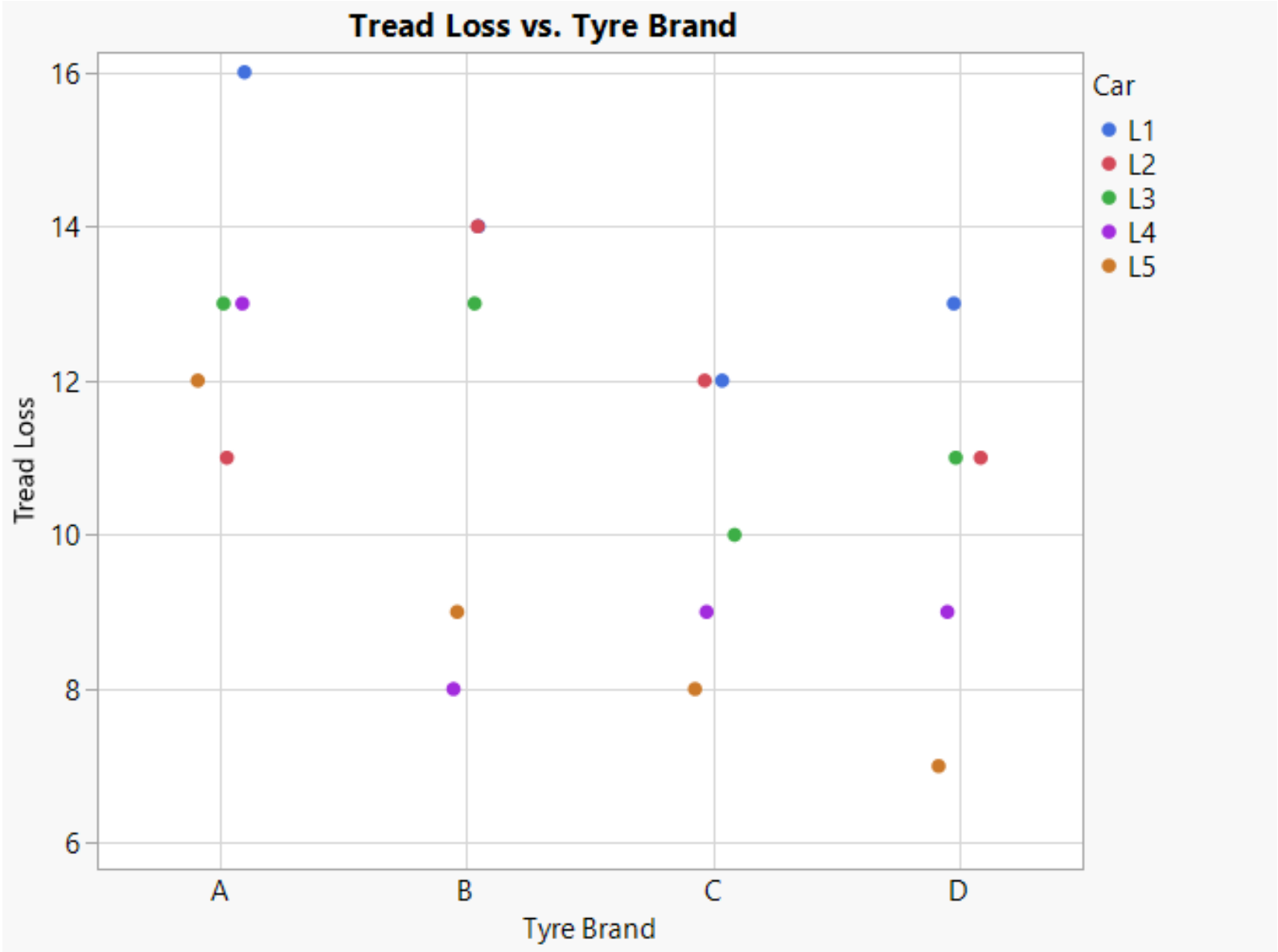
Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio	Prob > F
Tyre Brand	3	26.95000	8.98333	1.7789	0.1917
Error	16	80.80000	5.05000		
C. Total	19	107.75000			

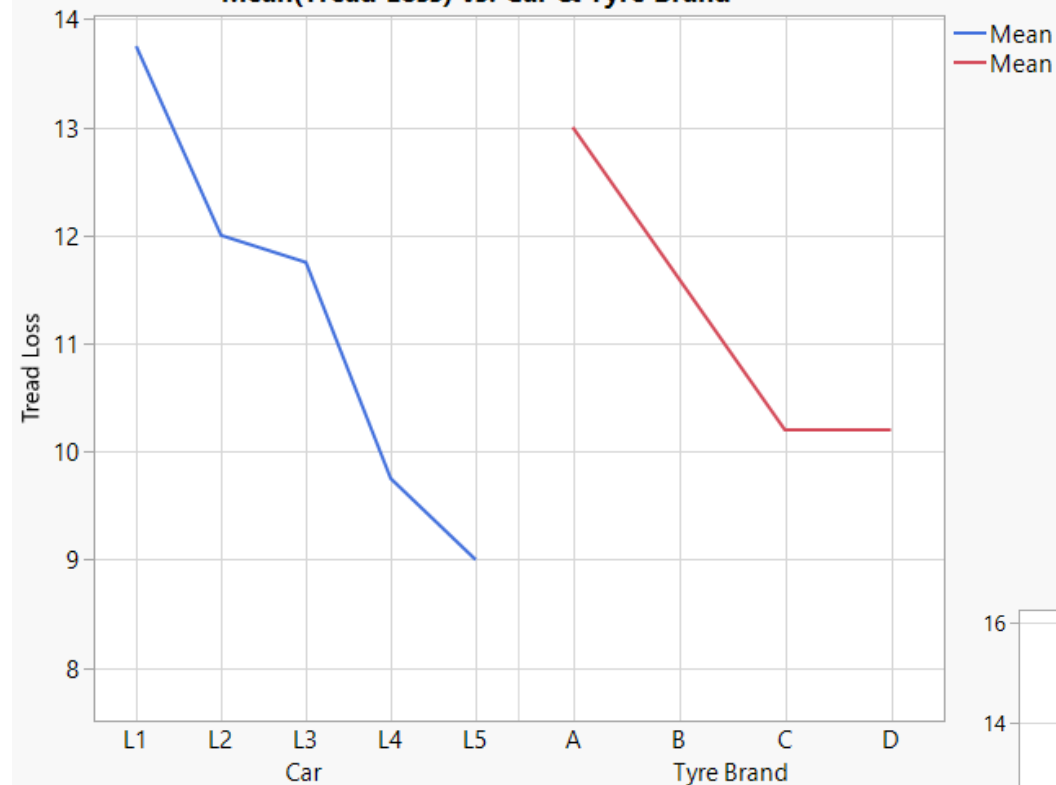
Conclusion? **The tyre brand is not significant**

Wrong Conclusion!

Someone suggested to plot the data explicitly considering the cars



Mean(Tread Loss) vs. Car & Tyre Brand

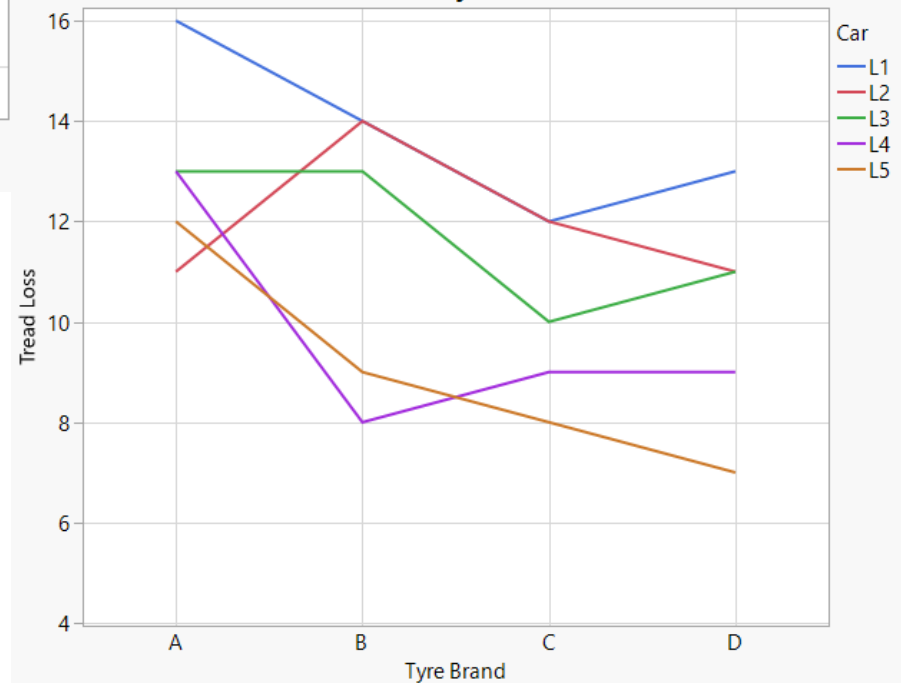


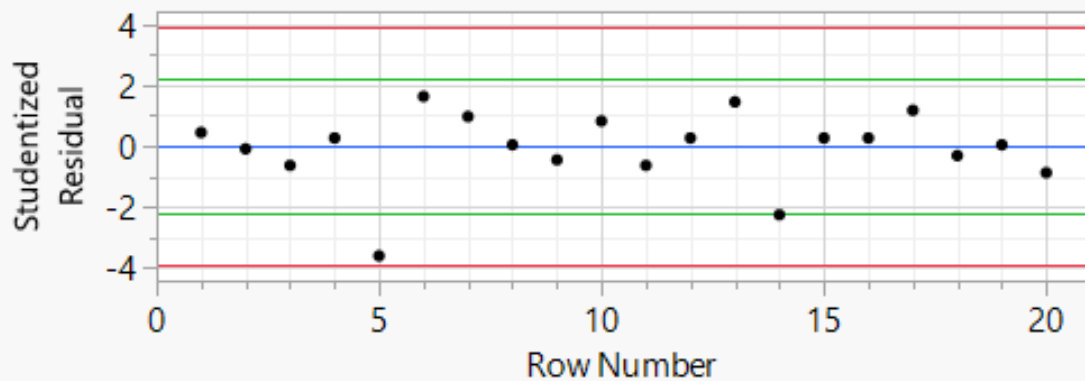
Let's try a new model

$$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon$$

Why «no interactions» in the model?

Tread Loss vs. Tyre Brand



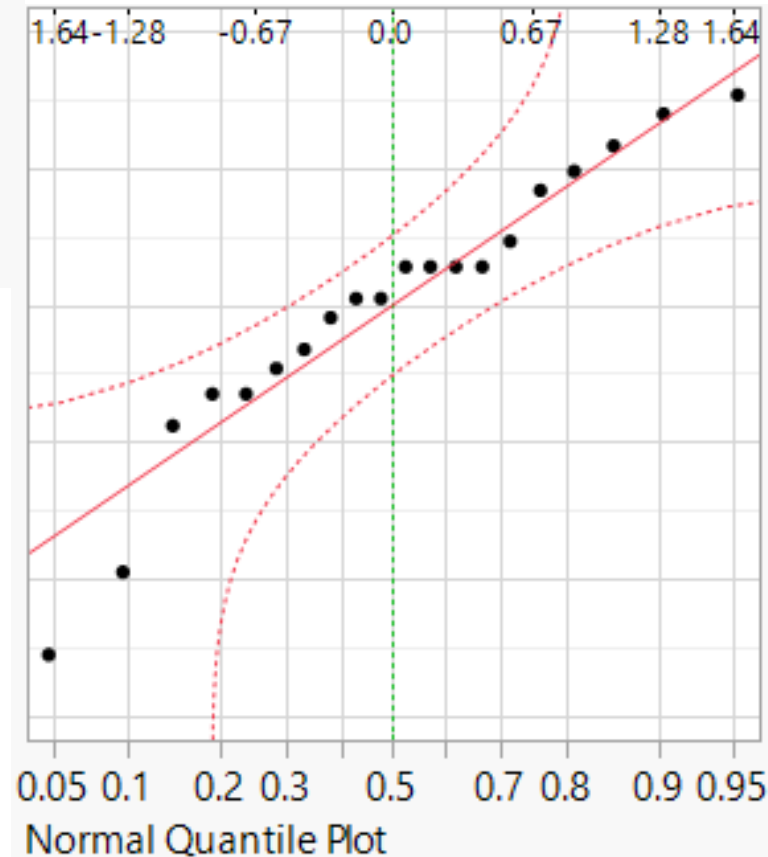


Goodness-of-Fit Test

	W	Prob<W
Shapiro-Wilk	0.9400367	0.2402
	A ²	Simulated p-Value
Anderson-Darling	0.4200329	0.3064

Note: H_0 = The data is from the Normal distribution.
Small p-values reject H_0 .

Hypothesis Checking



- We cannot check the equality of variances!
- We don't know the run order and we can not check the independency

ANOVA Table (2 factors, Additive Model)

Summary of Fit

RSquare	0.783759
RSquare Adj	0.657618
Root Mean Square Error	1.393437
Mean of Response	11.25
Observations (or Sum Wgts)	20

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	7	84.45000	12.0643	6.2134
Error	12	23.30000	1.9417	Prob > F
C. Total	19	107.75000		0.0031*

Effect Tests

Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
Car	4	4	57.500000	7.4034	0.0030*
Tyre Brand	3	3	26.950000	4.6266	0.0226*

From 10% to 66%

Conclusions?

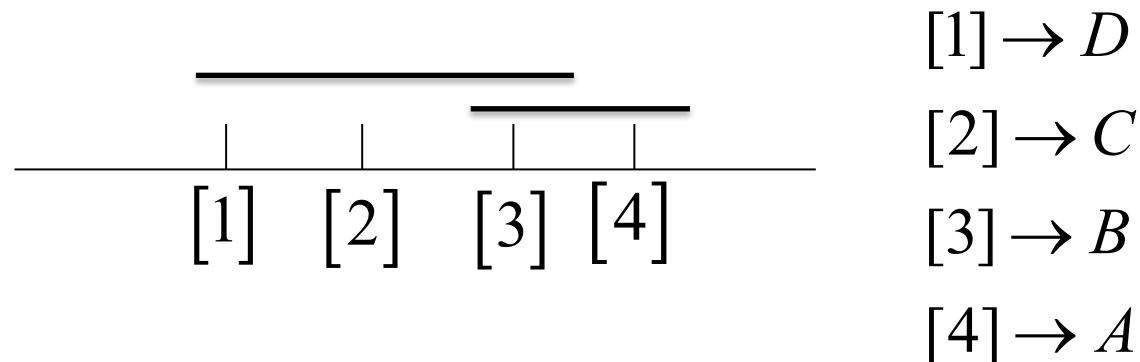
Brand Comparisons

Grouping Information Using the Tukey Method and 95% Confidence

Brand N		Mean	Grouping	
---------	--	------	----------	--

1	5	13.0	A	
2	5	11.6	A	B
3	5	10.2	B	
4	5	10.2	B	

Means that do not share a letter are significantly different.



However the **greatest influence** is due to cars: this would suggest a special car maintenance program

Recap: RCBD ANOVA Analysis

- Suppose a levels of the unique Factor of interest **A** and b levels of the Blocking Factor **B**
- Statistical model

$$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon$$

- For (**Fixed Effects**) we test:

$$H_0 : \tau_i = 0 \quad H_1 : \tau_i \neq 0 \text{ for at least one } i$$

$$\mu_i = \frac{1}{b} \sum_{j=1, b} (\mu + \tau_i + \beta_j) = \mu + \tau_i$$

$$H_0 : \mu_1 = \mu_2 = \dots \mu_a \quad H_1 : \mu_i \neq \mu_j \text{ for at least a couple } i, j$$

Multiple comparisons

Usually we are interested to compare the levels of the **Factor of interest**

Use the same methods previously seen with:

- Replace **b** for **n**
- The error degrees of freedom are $df_E = (\mathbf{a-1})(\mathbf{b-1})$

Additional remarks on RCBD

- RCBD uses an **additive model** – we assume no interaction and we use the **interaction mean square** to estimate the **error variance** MS_E
- Usually the analysis model should consider the Blocking Factor as a **Random Factor** (so the appropriate model should be a **Mixed model**)
- Next topics are:
 - **Missing** values?.
 - **Choice of the number of Blocks?**

Missing data (approximate analysis to have a Full & Balanced Table)

Fac	1	2	3	4
1	-2	-1	1	5
2	-1	-2	x	4
3	-3	-1	0	2
4	+2	+1	5	7

$\rightarrow y'_{2.}$

$y'_{.3}$ $y'_{..}$

$y'_{..}$ $y'_{l.}$ $y'_{.q}$

Totals **without** the missing x

- **Idea:** Assign to $\mathbf{x}$ the value that minimizes SS_E

$$\begin{aligned}
 SS_E &= \sum_{i,j} \left(y_{ij} - \hat{\mu} - \hat{\tau}_i - \hat{\beta}_j \right)^2 = \sum_{i,j} \left(y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot} \right)^2 = \\
 &= \sum_{i,j} y_{ij}^2 - \frac{1}{b} \sum_i \left(\sum_j y_{ij} \right)^2 - \frac{1}{a} \sum_j \left(\sum_i y_{ij} \right)^2 + \frac{1}{ab} \left(\sum_{i,j} y_{ij} \right)^2 = \\
 &= \mathbf{x}^2 - \frac{1}{b} \left(y'_{l\cdot} + \mathbf{x} \right)^2 - \frac{1}{a} \left(y'_{\cdot q} + \mathbf{x} \right)^2 + \frac{1}{ab} \left(y'_{\cdot\cdot} + \mathbf{x} \right)^2 + R
 \end{aligned}$$

in our case $l = 2$ $q = 3$

In $\mathbf{R}$ we have all terms that do not contain $\mathbf{x}$

$$y'_{2.} = 1 \quad y'_{.3} = 6 \quad y'_{..} = 17$$

$$a = 4 \quad b = 4$$

$$x = \frac{4 * 1 + 4 * 6 - 17}{3 * 3} = 1.22 = y_{23}$$

$$\frac{dSS_E}{dx} = 0 \quad x = \frac{ay'_{l.} + by'_{.q} - y'_{..}}{(a-1)(b-1)}$$

We make an ANOVA reducing by 1 the df_E and df_{TOT}

Source	SS	df	MS	F ₀	p-value
A	39.98	3	13.33	17.12	0.0008
B	79.53	3	26.51		
Error	6.22	8	0.78		It was 9
Total (Corrected)	125.73	14	It was 15		

Exact analysis (JMP->Fit Model)

Summary of Fit

RSquare	0.950513
RSquare Adj	0.913397
Root Mean Square Error	0.881917
Mean of Response	1.133333
Observations (or Sum Wgts)	15

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	6	119.51111	19.9185	25.6095
Error	8	6.22222	0.7778	Prob > F
C. Total	14	125.73333		<.0001*

Effect Tests

Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
A	3	3	39.527778	16.9405	0.0008*
B	3	3	78.944444	33.8333	<.0001*

$$39.53 + 78.94 = 118.47 \neq 119.51 \quad SS_A + SS_B \neq SS_{MOD}$$

We are using the **adjusted SS** since the experiment is not **Balanced**

Choice of the number of Blocks

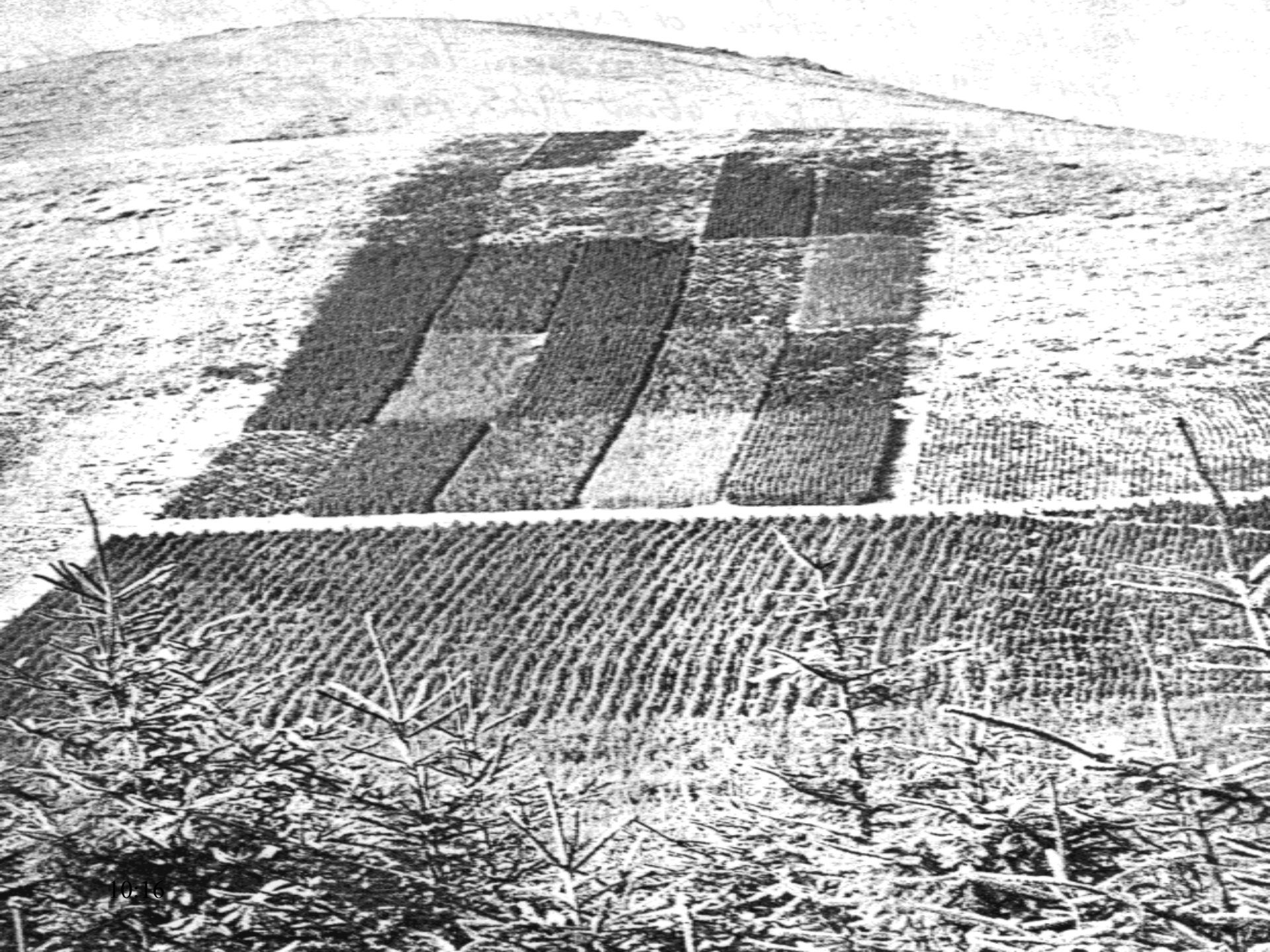
- We want to set the #levels of the blocking factor (**n=b**)
- We are interested in mean differences greater than **0.4**
- We know **$\sigma=0.1$** **a=4** and we set **alfa=0.05** and **Power ≥ 0.96**

$$\delta_A = \frac{b}{2} \left(\frac{d}{\sigma} \right)^2 \quad \text{for } n = 1 \quad \text{and} \quad \sum_{i=1,a} \tau_i^2 = \frac{d^2}{2}$$

$$\delta_A = 8b \quad df_{FAC} = \nu_1 = a - 1 \quad df_E = \nu_2 = (a - 1)(b - 1)$$

b	$df_A = a - 1$	$df_E = (a - 1)(b - 1)$	F_α	δ_A	Power
3	3	6	4.757	24	0.85
4	3	9	3.863	32	0.976
5	3	12	3.490	40	0.997





“Latin Square” plan

- **Latin Square plans** are used to handle **two Nuisance Factors** (there are extensions for more than two nuisance factors)
- **Model assumptions:** usual + the three factors (two nuisance and one main factor) **do not** interact.
- If the last assumption is violated, the Error Variance will be **overestimated** and it will be more difficult to recognize a significant factor
- As you can see in the photo, this kind of plans were extensively used in agricultural experiments.



This stained-glass window at **Gonville and Caius College**, Cambridge, honored R. Fisher. However the glass was **removed** in 2020 because of Fisher's connection with eugenic studies

- The number of levels for each Factor is p
- We set up a **square** matrix $p \times p$ of **latin letters**
- **Latin Letters** code **main factor levels**
- **Rows and columns** code the Blocking Factor levels
- For each specific couple of Blocking Factor levels we have just one Main Factor level (**Latin Letter**)

	<i>O1</i>	<i>O2</i>	<i>O3</i>
<i>P1</i>	<i>A</i>	<i>B</i>	<i>C</i>
<i>P2</i>	<i>B</i>	<i>C</i>	<i>A</i>
<i>P3</i>	<i>C</i>	<i>A</i>	<i>B</i>
- The total Number of runs is $N = p^2$
- **Note:** we have **three** factors and **three** levels but **N is not p^3**

If each entry of the square matrix is represented by a triple $\{r,c,s\}$, we can represent the square Latin in matrix form

	<i>O1</i>	<i>O2</i>	<i>O3</i>	$\begin{bmatrix} r & c & s \\ 1 & 1 & A \\ 1 & 2 & B \\ 1 & 3 & C \\ 2 & 1 & B \\ 2 & 2 & C \\ 2 & 3 & A \\ 3 & 1 & C \\ 3 & 2 & A \\ 3 & 3 & B \end{bmatrix}$	
<i>P1</i>	<i>A</i>	<i>B</i>	<i>C</i>		
<i>P2</i>	<i>B</i>	<i>C</i>	<i>A</i>		
<i>P3</i>	<i>C</i>	<i>A</i>	<i>B</i>		

A Latin Squares is a set of p^2 triples (r,c,s) where $1 \leq r,c,s \leq p$ such that all ordered pairs (r,c) , are distinct, all ordered pairs (r,s) are distinct and all ordered pairs (c,s) are distinct.

We have several Latin Square tables and we must randomly select one.

We do not have a **simple formula** for computing the number of Latin Square tables (L_p).

However, we have simple algorithms to get random Latin Squares.

Examples: $p=3$ $L_p=12$; $p=4$ $L_p=576$; $p=5$ $L_p=161280$;

Several Latin Squares (and L_p) are listed in **OEIS** the On-Line Encyclopedia of Integer Sequences

Try to represent the Treatments in a 3D space (two nuisance factors and one main Factor)

- **Statistical Model** (completely additive)

$$y_{ijk} = \mu + \tau_i + \beta_j + \alpha_k + \varepsilon_{ijk}$$

<i>Source</i>	<i>SS</i>	<i>df</i>	<i>MS</i>
Factor	$\sum_{i=1,p} \frac{y_{i..}^2}{p} - \frac{y_{...}^2}{p^2}$	$p-1$	MS_{FAC}
Block1	$\sum_{j=1,p} \frac{y_{.j.}^2}{p} - \frac{y_{...}^2}{p^2}$	$p-1$	MS_{BL1}
Block2	$\sum_{k=1,p} \frac{y_{..k}^2}{p} - \frac{y_{...}^2}{p^2}$	$p-1$	MS_{BL2}
Error	SS_E (by difference)	$(p-2)(p-1)$	MS_E
Total (corrected)	$\sum_{i,j,k} y_{ijk}^2 - \frac{y_{...}^2}{p^2}$	$p^2 - 1$	

Test $H_0 : \tau_i = 0$

$$\frac{MS_{FAC}}{MS_E} \sim F(p-1, (p-2)(p-1))$$

Example of computation

- Let us try to develop the first line of the ANOVA Table
- First we compute the model parameter estimates

$$\hat{\mu} = \frac{y_{...}}{p^2} \quad \longrightarrow \quad \hat{\tau}_i = \bar{y}_{i..} - \hat{\mu} = \frac{y_{i..}}{p} - \frac{y_{...}}{p^2}$$

$$\begin{aligned} SS_{FAC} &= \sum_{i,j,k} \hat{\tau}_i^2 = p \sum_i \hat{\tau}_i^2 = p \sum_i \left(\frac{y_{i..}}{p} - \hat{\mu} \right)^2 = p \sum_i \left(\frac{y_{i..}^2}{p^2} - 2\hat{\mu} \frac{y_{i..}}{p} + \hat{\mu}^2 \right) = \\ &= \sum_i \frac{y_{i..}^2}{p} - p 2\hat{\mu} p \hat{\mu} + p^2 \hat{\mu}^2 = \sum_i \frac{y_{i..}^2}{p} - p^2 \hat{\mu}^2 = \sum_i \frac{y_{i..}^2}{p} - \frac{y_{...}^2}{p^2} \end{aligned}$$

Missing data

- If one observation is missing we could substitute it

$$y_{lqk} = \frac{p(y'_{l..} + y'_{.q.} + y'_{..k}) - 2y'_{...}}{(p-2)(p-1)}$$

- Or we can analyze with a **regression approach** losing the orthogonality of the experiment

Example: pollution reduction

- The goal is to reduce the **Pollution** using 4 **Additives in gasoline (A, B, C, D)**
- Decide a typical **urban route**
- Nuisance Factors: **cars** e **drivers**
- Take some Cars (4) and some Drivers (4)
- Sample a specific Latin Square
- Sample the Cars and the Drivers
- Randomize the run order
- Perform the experiment

Car

<i>A</i>	<i>B</i>	<i>D</i>	<i>C</i>
<i>D</i>	<i>C</i>	<i>A</i>	<i>B</i>
<i>B</i>	<i>D</i>	<i>C</i>	<i>A</i>
<i>C</i>	<i>A</i>	<i>B</i>	<i>D</i>

Observation

$$A = 21 \quad B = 26 \quad D = 20 \quad C = 25$$

$$D = 23 \quad C = 26 \quad A = 20 \quad B = 27$$

$$B = 15 \quad D = 13 \quad C = 16 \quad A = 16$$

$$C = 17 \quad A = 15 \quad B = 20 \quad D = 20$$

Computations

$$y_{...} = 320 \quad \bar{y}_{...} = 20$$

Cars (columns)		Drivers (rows)		Additives	
$y_{.1.} = 76$	$\bar{y}_{.1.} = 19$	$y_{..I} = 92$	$\bar{y}_{..I} = 23$	$y_{A..} = 72$	$\bar{y}_{A..} = 18$
$y_{.2.} = 80$	$\bar{y}_{.2.} = 20$	$y_{..II} = 96$	$\bar{y}_{..II} = 24$	$y_{B..} = 88$	$\bar{y}_{B..} = 22$
$y_{.3.} = 76$	$\bar{y}_{.3.} = 19$	$y_{..III} = 60$	$\bar{y}_{..III} = 15$	$y_{C..} = 84$	$\bar{y}_{C..} = 21$
$y_{.4.} = 88$	$\bar{y}_{.4.} = 22$	$y_{..IV} = 72$	$\bar{y}_{..IV} = 18$	$y_{D..} = 76$	$\bar{y}_{D..} = 19$

ANOVA Computations

$$SS_{Additives} = \sum_{i=1,p} \frac{y_{i..}^2}{p} - \frac{y_{...}^2}{p^2} = \frac{72^2 + 88^2 + 84^2 + 76^2}{4} - \frac{320^2}{16} = 40$$

$$SS_{Car} = \sum_{j=1,p} \frac{y_{.j.}^2}{p} - \frac{y_{...}^2}{p^2} = \frac{76^2 + 80^2 + 76^2 + 88^2}{4} - \frac{320^2}{16} = 24$$

$$SS_{Drivers} = \sum_{k=1,p} \frac{y_{..k}^2}{p} - \frac{y_{...}^2}{p^2} = \frac{92^2 + 96^2 + 60^2 + 72^2}{4} - \frac{320^2}{16} = 216$$

$$SS_{Tot(corr)} = \sum_{i,j,k} y_{ijk}^2 - \frac{y_{...}^2}{p^2} = 6696 - \frac{320^2}{16} = 296$$

$$SS_E = SS_{Tot(corr)} - SS_{Additives} - SS_{Cars} - SS_{Drivers} = 16$$

- **ANOVA Table**

<i>Source</i>	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F₀</i>	
Additive	40	3	13. $\bar{3}$	5	$\rightarrow p\text{-value} = 4.5\%$
Driver	216	3	72.0	27	
Car	24	3	8.0	3	
Error	16	6	2.67		
Total (corrected)	296	15			

- **ANOVA Table without** considering the **Blocking Factors** (pay attention, this is a simple numerical example)

<i>Source</i>	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F₀</i>	<i>p - value</i>
Additive	40	3	13. $\bar{3}$	0.63	61.2%
Error	256	12	21.3		
Total (corr)	296	15			

Replicates

- When we have few levels of p , the error degrees of freedom $(p-2)(p-1)$ to estimate σ^2 (through MS_E) may not be enough i.e. if $p=3$ $df_E=2$
- We can replicate the experiment n times
- Possibilities are
 - A) Use the same block levels (same drivers and cars)
 - B) Change the levels of one block only
 - C) Change the levels of both blocks

ANOVA Table **Case A** (same levels)

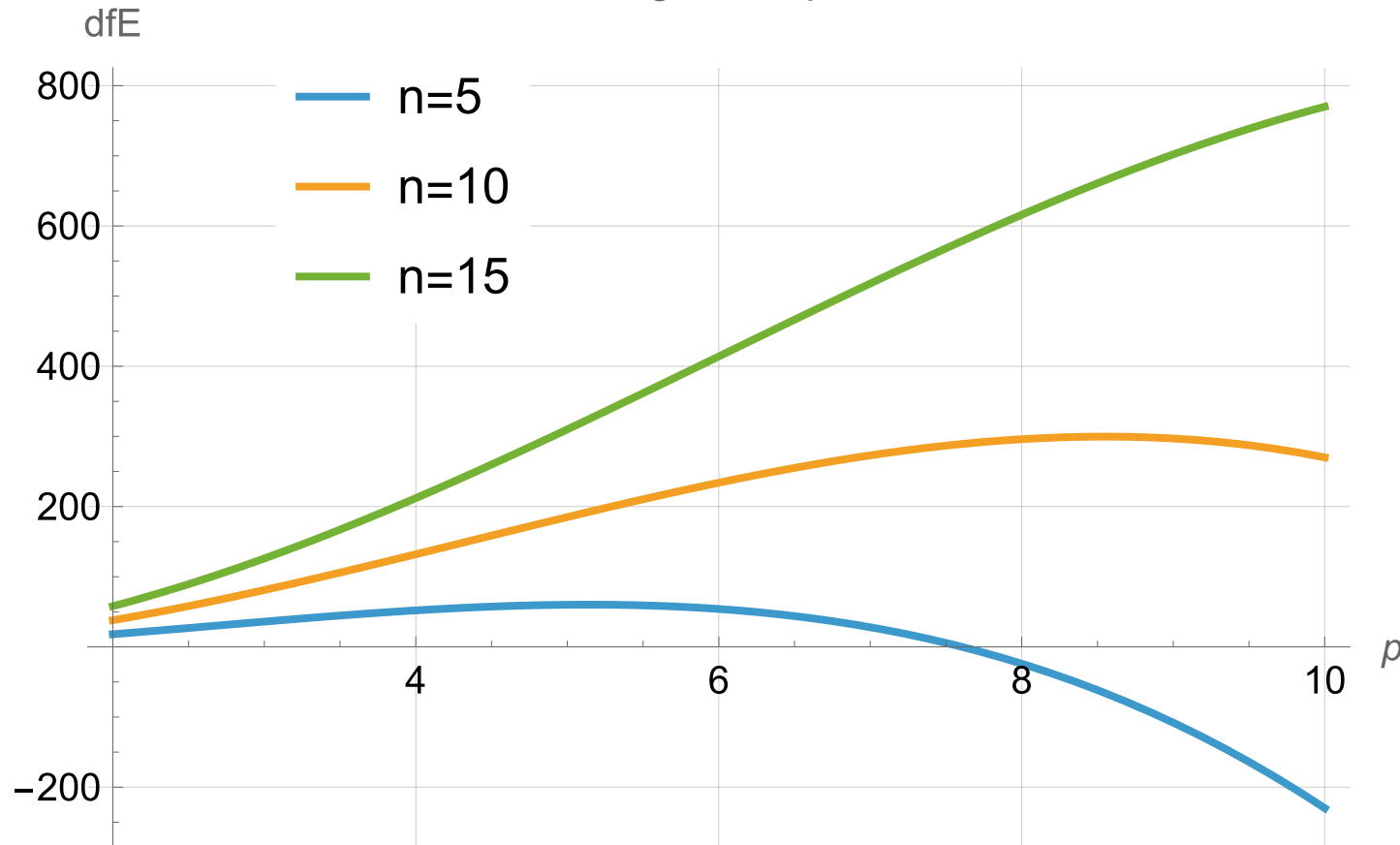
Source	<i>SS</i>	<i>df</i>	<i>MS</i>
Factor	$\sum_{i=1,p} \frac{y_{i...}^2}{np} - \frac{y_{...}^2}{np^2}$	$p - 1$	MS_{FAC}
Block1	$\sum_{j=1,p} \frac{y_{.j..}^2}{np} - \frac{y_{...}^2}{np^2}$	$p - 1$	MS_{BL1}
Block2	$\sum_{k=1,p} \frac{y_{..k.}^2}{np} - \frac{y_{...}^2}{np^2}$	$p - 1$	MS_{BL2}
Replicates	$\sum_{l=1,n} \frac{y_{...l}^2}{p^2} - \frac{y_{...}^2}{np^2}$	$n - 1$	MS_{REP}
Error	by difference	$(p - 1)[n(p + 1) - 3]$	MS_E
Total (Corr)	$\sum_{i,j,k,l} y_{ijkl}^2 - \frac{y_{...}^2}{np^2}$	$np^2 - 1$	

Some analysts do not consider replications in the error term since they were done in different moments

- If we consider the replication contribution in the Error term

$$df_E = np^2 - 1 - (p-1)^3$$

dfE without eliminating the replication contribution



ANOVA table **Case B** (change the levels of **Block1**)

Source	<i>SS</i>	<i>df</i>	<i>MS</i>
Factor	$\sum_{i=1,p} \frac{y_{i...}^2}{np} - \frac{y_{...}^2}{np^2}$	$p - 1$	MS_{FAC}
Block1	$\sum_{\substack{j=1,p \\ l=1,n}} \frac{y_{\cdot j \cdot l}^2}{p} - \sum_{l=1,n} \frac{y_{...l}^2}{p^2}$	$n(p - 1)$	MS_{BL1}
Block2	$\sum_{k=1,p} \frac{y_{\cdot \cdot k \cdot}^2}{np} - \frac{y_{...}^2}{np^2}$	$p - 1$	MS_{BL2}
Replicates	$\sum_{l=1,n} \frac{y_{...l}^2}{p^2} - \frac{y_{...}^2}{np^2}$	$n - 1$	MS_{REP}
Error	by difference	$(p - 1)[np - 2]$	MS_E
Total (Corr)	$\sum_{i,j,k,l} y_{ijkl}^2 - \frac{y_{...}^2}{np^2}$	$np^2 - 1$	

Application example: Crossover Design

- We are interested in analysing the efficacy of $p=2$ medicines (A, B), but people variability could mask the medicine efficacy (**first nuisance factor: people**).
- We use the same person in two distinct moments to avoid the possible chemical interaction of the two medicines on the same person (**second nuisance factor: time of supply**).
- Summing up: we conduct the experiment in $p=2$ distinct moments using a total of $np=20$ persons.
- In a first step we give the medicine A to a group of $np/2$ persons and the medicine B to the others. In the second step, after an appropriate period of time, we **cross** the experimentation between the two groups of people.

- Basically we have **10** “Latin Squares” plans, the first blocking factor is changed (the person), the second one (the time) is NOT changed

Person →	1	2	Person →	3	4	Person →	5	6
Time1	A	B	Time1	B	A	Time1	A	B
Time2	B	A	Time2	A	B	Time2	B	A

Person →	7	8	Person →	9	10	Person →	11	12
Time1	A	B	Time1	A	B	Time1	B	A
Time2	B	A	Time2	B	A	Time2	A	B

.....

Person →	19	20
Time1	A	B
Time2	B	A

Case B

• **ANOVA Table Case C** (All levels are changed)

<i>Source</i>	<i>SS</i>	<i>df</i>	<i>MS</i>
Factor	$\sum_{i=1,p} \frac{y_{i...}^2}{np} - \frac{y_{...}^2}{np^2}$	$p - 1$	MS_{FAC}
Block1	$\sum_{\substack{j=1,p \\ l=1,n}} \frac{y_{.j.l}^2}{p} - \sum_{l=1,n} \frac{y_{...l}^2}{p^2}$	$n(p - 1)$	MS_{BL1}
Block2	$\sum_{\substack{k=1,p \\ l=1,n}} \frac{y_{..kl}^2}{p} - \sum_{l=1,n} \frac{y_{...l}^2}{p^2}$	$n(p - 1)$	MS_{BL2}
Replicates	$\sum_{l=1,n} \frac{y_{...l}^2}{p^2} - \frac{y_{...}^2}{np^2}$	$n - 1$	MS_{REP}
Error	By subtraction	$(p - 1)[n(p - 1) - 1]$	MS_E
Total (Corr)	$\sum_{i,j,k,l} y_{ijkl}^2 - \frac{y_{...}^2}{np^2}$	$np^2 - 1$	

Graeco-Latin Square Design

- Graeco-Latin Squares are used when we need to block **three** nuisance factors
- The number of levels are p for each factor and the number of experiments is p^2

$A\alpha$	$B\beta$	$C\gamma$	$D\delta$
$B\delta$	$A\gamma$	$D\beta$	$C\alpha$
$C\beta$	$D\alpha$	$A\delta$	$B\gamma$
$D\gamma$	$C\delta$	$B\alpha$	$A\beta$

Model (Completely additive)

$$y_{ijkl} = \mu + \tau_i + \theta_j + \omega_k + \varphi_l + \varepsilon_{ijkl}$$

- The **time length** of a chemical reaction depends on the type of **Catalyst** used. (we have $a=4$ **different catalysts**)
- The **reaction time** depends, probably, in some but unknown way, on the **material batch** used (**Nuisance factor**)
- The material batch is sufficient to run only $k=3$ Catalysts per batch ($k < a$)
- In this situation we could use a

BIBD- Balanced Incomplete Block Design

- **Each pair** of levels of the **Factor (Catalyst)** appears together an equal number of times λ through all the b **blocks**
- **Example 1** on the board:
 - $a=4$ level factor, $k=3$ run per block
 - $a=4$ $k=2$

I need $b = \binom{a}{k}$ Blocks

In our example $\mathbf{b} = 4!/(3!1!)=4$

Catalyst ↓	1	2	3	4	$y_{i\cdot}$ ↓
1	73	74	—	71	218
2	—	75	67	72	214
3	73	75	68	—	216
4	75	—	72	75	222
$y_{\cdot j} \rightarrow$	221	224	207	218	870

Statistical Model $y_{ij} = \mu + \tau_i + \beta_j + \varepsilon$

- The experiment is described by the following matrix

		Block (j)					
		1	2	...	j	...	b
Factor (i).. i .. a	1	δ_{11}	δ_{12}	..	δ_{1j}	..	δ_{1b}
	2	δ_{21}	δ_{22}	..	δ_{2j}	..	δ_{2b}
	..	..	..	..	..	..	..
	i	δ_{i1}	δ_{i2}	..	δ_{ij}	..	δ_{ib}
	.. a	.. δ_{a1}	.. δ_{a2}		.. δ_{aj}		.. δ_{ab}

$$\delta_{ij} = \begin{cases} 1 & \text{if level } i \in B_j \\ 0 & \text{if level } i \notin B_j \end{cases}$$

$$\sum_{\substack{i=1,a \\ j=1,b}} \delta_{ij} = N$$

$$\sum_{i=1,a} \delta_{ij} = k$$

Total number of runs per block $\forall j$

$$\sum_{j=1,b} \delta_{ij} = r$$

Number of times we observe level i

$$\forall (i, i') \rightarrow \sum_{j=1,b} \delta_{ij} \delta_{i'j} = \lambda$$

$$N = ra = kb$$

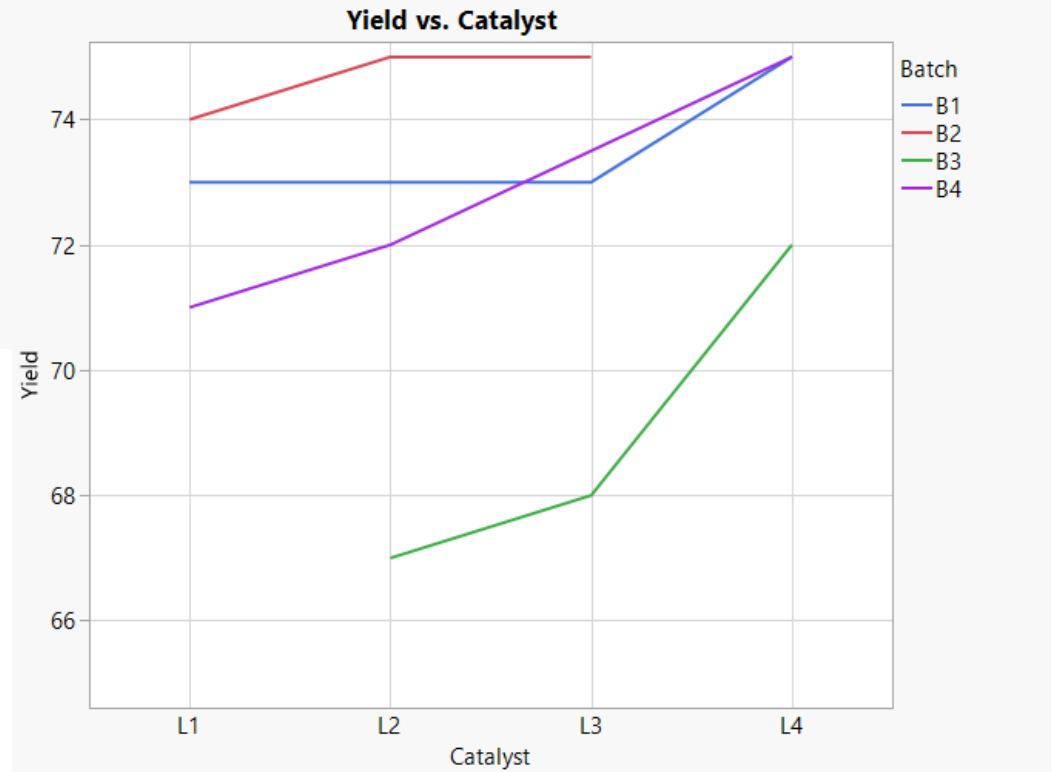
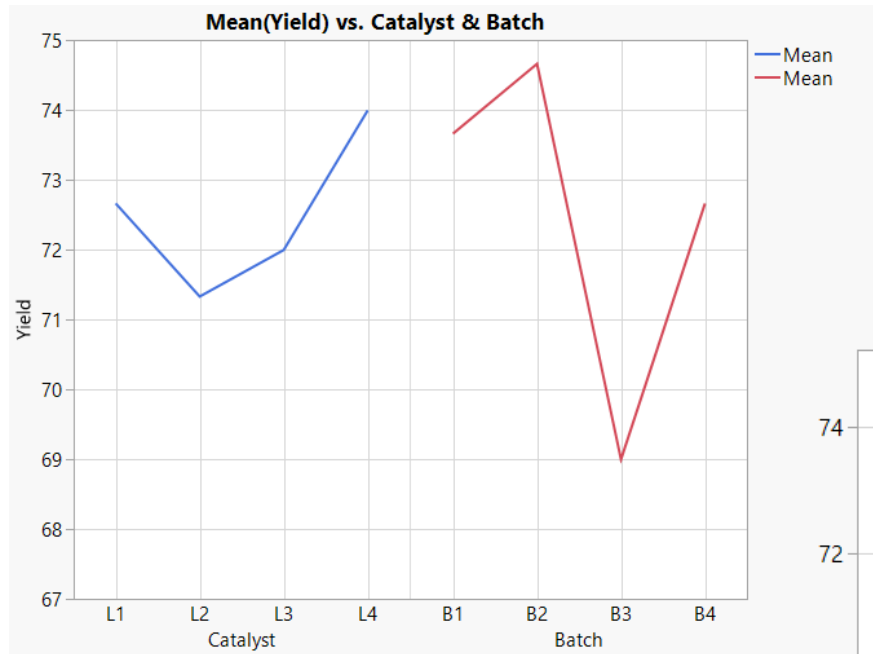
$$r = \frac{kb}{a} = \frac{k}{a} \binom{a}{k} = \frac{(a-1)!}{(k-1)!(a-k)!}$$

- a main factor number of levels
- b number of Blocks
- k number of runs in a Block
- N total number of experiments $N=bk$
- r number of times (replicates) each level occurs in the design $N=bk=ar \rightarrow r=N/a$
- λ number of times **each pair** of levels appears together (across the blocks)

$$\forall i \rightarrow \begin{cases} \text{number of pairs} = r(k-1) \\ \text{considering the levels} \rightarrow \lambda(a-1) \end{cases} \quad r(k-1) = \lambda(a-1)$$

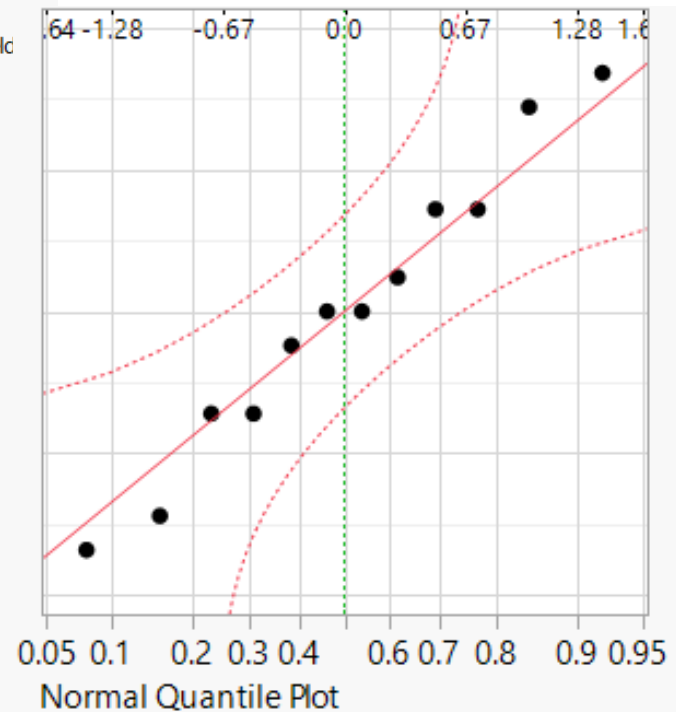
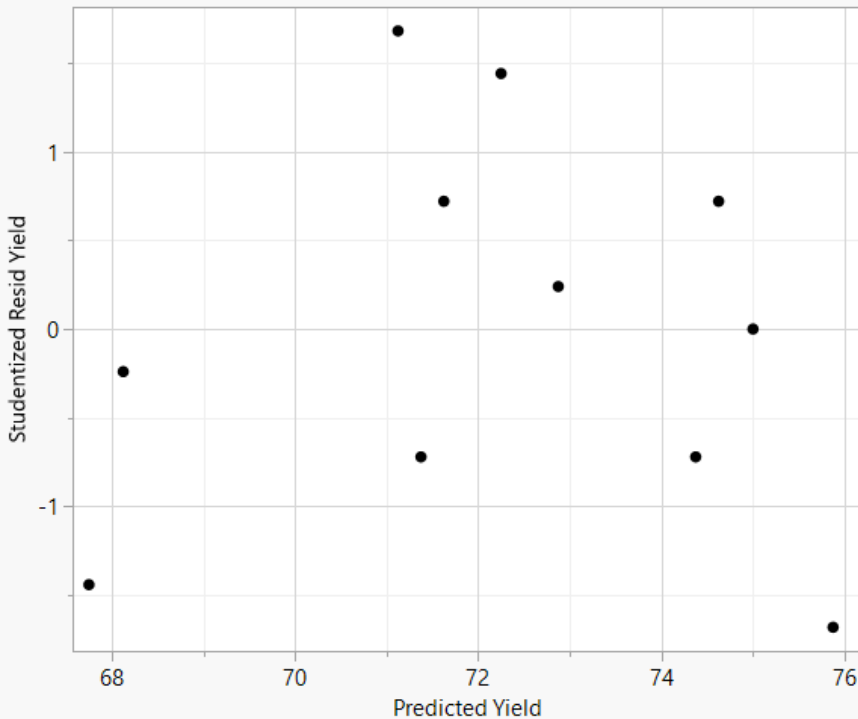
$$\lambda = \frac{r(k-1)}{a-1} = \frac{(a-2)!}{(a-k)!(k-2)!} = \frac{3(3-1)}{4-1} = 2$$

Usual preliminary graphs



Statistical Model $y_{ij} = \mu + \tau_i + \beta_j + \varepsilon$

Studentized Resid Yield vs. Predicted Yield



Residual analysis

Goodness-of-Fit Test

	W	Prob<W
Shapiro-Wilk	0.9694547	0.9050
	A ²	Simulated p-Value
Anderson-Darling	0.1726502	0.9228

Note: Ho = The data is from the Normal distribution.
Small p-values reject Ho.

ANOVA

Summary of Fit

RSquare	0.959877
RSquare Adj	0.911728
Root Mean Square Error	0.806226
Mean of Response	72.5
Observations (or Sum Wgts)	12

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	6	77.750000	12.9583	19.9359
Error	5	3.250000	0.6500	Prob > F
C. Total	11	81.000000		0.0024*

Effect Tests

Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
Catalyst	3	3	22.750000	11.6667	0.0107*
Batch	3	3	66.083333	33.8889	0.0010*

All Pairwise Differences Connecting Letters

Catalyst		Least Squares Mean
L4	A	75.000000
L3	B	72.000000
L2	B	71.625000
L1	B	71.375000

Levels not connected by same letter are significantly different.

In section 4.4 (Monty) there are details on estimates and SSs formula

a	k	b	r	λ	N	df_E	
3	2	3	2	1	6	1	$b = \begin{pmatrix} a \\ k \end{pmatrix}$
4	3	4	3	2	12	5	
4	2	6	3	1	12	3	$N = bk = ar$
5	4	5	4	3	20	11	
5	3	10	6	3	30	16	$r = bk / a$
5	2	10	4	1	20	6	
6	5	6	5	4	30	19	$\lambda = \frac{r(k-1)}{a-1}$
6	4	15	10	6	60	40	
6	3	20	10	4	60	35	$df_E = (N-1) - (a-1) - (b-1)$
6	2	15	5	1	30	10	
7	6	7	6	5	42	29	$= N - a - b + 1$
7	5	21	15	10	105	78	
7	4	35	20	10	140	99	
7	3	35	15	5	105	64	
7	2	21	6	1	42	15	

Watch the jumps for N

Try listing yourself the Blocks for 5-level factor, k=3 runs per block

Blocking in Factorial Design

- Let's consider a factorial design with two factors (**A**, **B**) and *n* replicates
- The total number of experiments is $N=nab$

If we can guarantee homogeneous conditions only *ab* runs, we can randomize inside the *ab* runs and analyse using:

$$y_{ijk} = \mu + \tau_i + \beta_j + \tau\beta_{ij} + \delta_k + \varepsilon$$

δ_k Block effect we are assuming no interaction between factor and blocking factor. If the interaction exists, the effect of ignoring it would be to overestimate the error variance.

<i>Source</i>	<i>SS</i>	<i>df</i>	<i>E(MS)</i>
Blocks	$\frac{1}{ab} \sum_k y_{..k}^2 - \frac{y_{...}^2}{abn}$	$n - 1$	$\sigma^2 + ab\sigma_\delta^2$ or $\sigma^2 + \frac{ba}{n-1} \sum_k \delta_k^2$
A	$\frac{1}{bn} \sum_i y_{i..}^2 - \frac{y_{...}^2}{abn}$	$a - 1$	$\sigma^2 + \frac{bn}{a-1} \sum_i \tau_i^2$
B	$\frac{1}{an} \sum_j y_{.j.}^2 - \frac{y_{...}^2}{abn}$	$b - 1$	$\sigma^2 + \frac{an}{b-1} \sum_j \beta_j^2$
AB	$\frac{1}{n} \sum_{i,j} y_{ij.}^2 - \frac{y_{...}^2}{abn} - SS_A - SS_B$	$(a-1)(b-1)$	$\sigma^2 + \frac{n \sum_{i,j} \tau \beta_{ij}^2}{(a-1)(b-1)}$
Error	by subtraction	$(ab-1)(n-1)$	σ^2
Total	$\sum_{i,j,k} y_{ijk}^2 - \frac{y_{...}^2}{abn}$	$abn - 1$	